Complete schematic - sheet 4

Every component terminal is shown. Equal net labels are connected; use connection IDs in the CSV for physical routing.

E1 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 02 Nov 2026

US_R | HC-SR04 right

INVENTORY / VERIFY

module terminal symbol

VCC	V5_LOGIC
TRIG	TR_5V
ECHO	ER_RAW
GND	GND

M5 | brush 1 motor

INVENTORY / VERIFY

12 V

A	M	M5_A
B		DRV_A_OUT2

CAM1 | ESP32-CAM - later phase

INVENTORY / VERIFY

reserved terminal symbol

5V	NC - INSULATE
GND	NC - INSULATE
U0R	NC - INSULATE
U0T	NC - INSULATE

FM3 | motor 3 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	DRV_R_MP
OUT	M3_A

FM6 | motor 6 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	DRV_A_OUT1
OUT	M6_A

R24 | ECHO lower resistor

ADDITIONAL / MISSING OR SELECT

3.3k 1%, >=1/8 W

1	EF_SAFE
2	GND

M1 | left front wheel motor

INVENTORY / VERIFY

12 V / 30 RPM

A	M	M1_A
B		DRV_L_MN

M2 | left rear wheel motor

INVENTORY / VERIFY

12 V / 30 RPM

A	M	M2_A
B		DRV_L_MN

M3 | right front wheel motor

INVENTORY / VERIFY

12 V / 30 RPM

A	M	M3_A
B		DRV_R_MN

M4 | right rear wheel motor

INVENTORY / VERIFY

12 V / 30 RPM

A	M	M4_A
B		DRV_R_MN

M6 | brush 2 motor

INVENTORY / VERIFY

12 V

A	M	M6_A
B		DRV_A_OUT2

M7 | conveyor motor

INVENTORY / VERIFY

Voltage TBD

A	M	M7_A
B		DRV_A_OUT4

FM1 | motor 1 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	DRV_L_MP
OUT	M1_A

FM2 | motor 2 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	DRV_L_MP
OUT	M2_A

FM4 | motor 4 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	DRV_R_MP
OUT	M4_A

FM5 | motor 5 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	DRV_A_OUT1
OUT	M5_A

FM7 | motor 7 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	DRV_A_OUT3
OUT	M7_A

R23 | ECHO upper resistor

ADDITIONAL / MISSING OR SELECT

2.2k 1%, >=1/8 W

1	EF_RAW
2	EF_SAFE

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.